INCH-POUND

MIL-PRF-52955B
23 December 1996
SUPERSEDING
MIL-P-52955A
19 July 1984

PERFORMANCE SPECIFICATION
POSITION AND AZIMUTH DETERMINING SYSTEM (PADS): AN/USQ-70

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an inertial surveying system that is air and ground vehicle mounted.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in section 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. ARMY TOPOGRAPHIC ENGINEERING CENTER, ATTN CETEC-PD-TE, 7701 TELEGRAPH ROAD, ALEXANDRIA, VA 22315-3864 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6675

Distribution Statement A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE

MIL-STD-462 - Measurement of Electromagnetic Interference Characteristics

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AERONAUTICAL RADIO, INCORPORATED

ARINC 607-1 - Design Guidance for Avionic Equipment.

(Application for copies should be addressed to Aeronautical Radio, Incorporated, 2551 Riva Road, Mail Stop 5-123, Annapolis, MD 21401.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electric Code

(Application for copies should be addressed to the National Fire Protection Association International, 60 Batterymarch Park, Quincy, MA 02269-9101.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC.

SAE ARP 1199 - Selection, Application, and Inspection of Electric Overcurrent Protective Devices (DoD Adopted).

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

UNDERWRITER'S LABORATORIES INC.

UL 943 - Underwriters Laboratories Inc., Standards Service for Ground-Fault Circuit-Interrupters.

(Application for copies should be addressed to Underwriter's Laboratories Inc., 333 Pfingston Road, Northbrook, IL 60062-2096.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The Position and Azimuth Determining System (hereinafter called "PADS") shall provide three dimensional coordinates and azimuth. PADS shall be capable of rapid and accurate self-alignment independent of the global positioning system (GPS). The PADS shall be configured as a self-contained unit, as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.3 Interface requirements. When installed in a host air/ground vehicle, the PADS shall operate without interference with or from the normal operation of the host vehicle or the vehicle accessories. The PADS shall be capable of being mounted in all host vehicles, see 3.3.1 and 3.3.2, utilizing additional framework, if required. No permanent modification to the host vehicle is permitted, with the exception of replacement hardware. This interface shall be provided as required (see 6.2).

3.3.1 Interface with ground vehicle. The PADS shall interface with the following as host ground vehicles (See 3.17):

a. Utility Truck: Cargo/Troop carrier, 1-1/4 ton, 4 x 4, M998, High Mobility Multi-Wheeled Vehicle (HMMWV).

b. Commercial Utility Cargo Vehicle (CUCV), M1009.

c. Carrier, cargo, 1-1/2 Ton, M973 Small Unit Support Vehicle (SUSV).

3.3.2 Interface with air vehicle. The PADS shall interface with the following as host air vehicles (See 3.17):

a. Aircraft; Rotary Wing, Observation: OH-58.

b. Aircraft; Rotary Wing, Utility: UH-1 and UH-60.

c. Aircraft; Rotary Wing, Cargo-Transport: CH-46, CH-47 and CH-53.

3.4 Major component characteristics.

3.4.1 Measurement and computation unit(s). The measurement and computation unit(s) shall establish and maintain the orientation of a coordinate reference frame to which all measurements are referred. The device(s) shall perform all computations required to continually provide such survey parameters as horizontal position in Universal Transverse Mercator (UTM), geographic coordinates, elevation in meters, and azimuth in mils. The device shall also:

a. Maintain the coordinate references;

b. Control system errors;

c. Accomplish Spheroid and UTM coordinate conversions;

d. Condition signals;

e. Monitor system performance;

f. Control gyrocompassing.

3.4.1.1 Selectability. The PADS shall have a spheroid database capable of handling not less than sixteen data sets, fourteen predetermined spheroids and two positions for user defined spheroids. The operator shall be able to select the desired spheroid for use at the beginning of the survey mission, with operator entries in decimal numbers via the CDU. The fourteen predetermined spheroids, and two supplementary positions, shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

a. Clark 1866
b. International
c. Clarke 1880
d. Everest
e. Bessel
f. Spare
g. Spare
h. Australian National/South American
i. Geodetic Reference System 1980, WGS-84
j. Airy
k. Modified Airy
l. Modified Everest
m. World Geodetic System (WGS-72)
n. Hough
o. User defined
p. User defined

3.4.2 Control and Display Unit (CDU). The CDU shall provide the communications interface between the PADS and the operator. From a single location, the operator shall be able to control the PADS mode of operation; accomplish data entry, readout, and update; observe systems status; and display survey and other operational parameters. The CDU shall have a screen to display Easting, Northing, Height and Azimuth. A visual display shall provide the operator with update and malfunction status. Three-dimensional coordinates with resolution to one-tenth of a meter shall be provided for horizontal position (Northing and Easting) and elevation, and 0.001 arc second in displaying latitude and longitude. The UTM zone and hemisphere of operation shall be displayed. The grid azimuth reference and true azimuth shall be displayed with resolution to one-hundredth of a mil. Provision shall be made for entering the initial three-dimension coordinates. When operating within 80 kilometers of an adjacent UTM grid zone, provision shall be made to display the adjacent zone position coordinates and grid azimuth. A CDU extension cable at least six feet in length (to be used from the operator position) shall provide an interface for the electrical connections from the PADS computer to the CDU. The extension cable shall be stored within the PADS configuration when not in use.

3.4.2.1 CDU Operational characteristics. The CDU shall be capable of being remotely located from the PADS. During the PADS operation in the host vehicle, the CDU shall be adjustable in a position where it is visible and accessible for hands-free operation (except for data entry and read-out procedures) by both the vehicle driver and PADS operator (not necessarily at the same time). The CDU shall be as specified herein without modification to the host vehicle.

3.4.3 Device for offset coordinate and azimuth transfer. Means shall be provided to transfer position coordinates in and out of PADS, from and to a location offset from the host vehicle. Means shall be provided to transfer azimuth from PADS to an orienting line, offset from the host vehicle (see 3.10.3).

3.4.4 Security device. The PADS shall provide a means to delete all entries in the survey control point file (commonly called the mark table).

3.4.5 Vibration and shock isolators. Vibration and shock isolators, if used, shall be constructed so that failure of the resilient material will not free the supported equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

3.4.6 Back-up power source. The PADS electrical power system shall include a back-up power source (battery) which will sustain the operation of the PADS after loss of host vehicle power (see 3.11.2). The back-up power source shall operate the PADS for not less than 15 minutes after loss of external power. Loss of external power shall not damage the system or permit loss of data.

3.4.6.1 Battery selection. It is preferred that battery(ies) acquired for use in the PADS be a DoD approved battery.

3.4.6.2 Battery safety provisions. The battery unit shall be provided with devices to firmly secure the battery(ies). Adequate room shall be provided for battery installation, maintenance, testing and removal without disassembly of the equipment. The housing shall prevent pressure build-up from heat, gases, rupture, or chemicals released during battery operation and charging, within the enclosure and shall also prevent such materials from entering the electronic compartment.

3.4.7 Cables. Power interface cables and adapters, as required, shall be provided for the specific host vehicle(s) (see 6.2). Means shall be provided for their storage in or on the PADS, when not in use.

3.4.8 Transit case. A transit case shall be provided with the PADS. The case shall be constructed of materials selected by the contractor, as specified herein. The interior of the PADS transit case shall include provisions to securely mount the PADS. The transit case shall not sustain damage to an extent that would cause dysfunction of the enclosure hinges or the mounting provisions inside the case, when tested as specified in 4.6.6.

3.4.8.1 Durability. When tested as specified in 4.6.6.1, the PADS transit case shall show no evidence of damaged seals, welds that impair structural integrity or malfunctioning latches, carrying devices, and pressure relief valves. Any dents or warpage sustained shall not impair normal opening and closing of the case.

3.4.8.2 Survivability. The transit cases shall be subjected to the free-fall drop test specified in 4.6.6.2. The cases shall show no evidence of damaged seals, welds that impair structural integrity or malfunctioning latches, carrying devices, and pressure relief valves. Any dents or warpage sustained shall not impair normal opening and closing of the case.

3.4.8.3 Leakage. The transit cases shall not permit excessive leakage when tested in accordance with 4.6.6.3. Water penetration of not more than 4 cubic centimeters per 28,000 cubic centimeters shall be acceptable.

3.4.9 Accessory storage. Storage shall be provided within the PADS for all contractor supplied manuals, tools, accessories, cables and supplies.

3.4.10 Tool kit. A tool kit shall be included with the accessory items. All tools necessary to perform routine maintenance and to install the PADS in the host vehicles shall be supplied. The tools shall be provided in a durable fabric or leather pouch.

3.4.11 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable (see 6.7.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

3.5 Construction.

3.5.1 Built-in test equipment (BITE). The PADS shall contain BITE that will notify the operator of marginal performance, equipment malfunction, and operator error. The BITE shall isolate a malfunction down to the major component level.

3.6 Reliability. The PADS shall demonstrate a mean-time between-failure (MTBF) of at least 2000 hours when tested as specified in 4.6.1.2.

3.6.1 Burn-in. Each PADS shall be subjected to procedures and/or processes to weed out workmanship problems and infant parts mortality. Procedures and/or processes shall be submitted to the Government for approval, otherwise each PADS shall be subjected to the burn-in procedure specified in 4.5.1.1.

3.7 Maintainability.

3.7.1 Maintenance test equipment. Maintenance test equipment shall be supplied, as specified in 6.2.

3.7.2 Maintenance support. Technical manuals, repair parts, and tools supplied shall be capable of performing intended functions of assisting in or conducting maintenance operations, when tested as specified in 4.6.1.3.1.

3.7.3 Maintenance ratio. The PADS shall have a maintenance ratio of not more than 0.02 (see 6.7.2). Normal before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.8 Safety. The PADS shall not present uncontrolled safety hazards (see 6.7.3).

3.8.1 Electrical connections. A mistake-proof means to prevent the inadvertent reversing or mismating of electrical connections shall be provided.

3.8.2 Battery installation markings. Connections, polarity, normal voltage, minimum acceptable voltage for equipment operation, and the type of batteries required shall be marked in a prominent place on or near the battery unit.

3.8.3 Caution label. The PADS shall be labeled externally as follows:

CAUTION
REMOVE BATTERIES BEFORE
SHIPMENT
OR
INACTIVE STORAGE OF 30 DAYS OR MORE

3.8.4 Transit cases. Transit cases shall be provided with an automatic pressure relief valve.

3.8.5 Protection devices. The PADS electrical circuitry shall be protected from damage by electrical surges and spikes, using standard current interrupt devices, for use in air and ground vehicles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

3.9 Noise limits. The PADS, at any stage of operation, shall not emit sound interference sufficient to disrupt voice communication, by radio or telephone, when verified as specified in 4.6.3.

3.10 Performance characteristics.

3.10.1 Vehicle dynamics. The PADS shall be required to perform as specified herein when operated within the following envelopes for vehicle dynamics:

a. Horizontal velocity, ground speed: 0 to 150 knots, in any direction.

b. Vertical velocity: 0 to 2400 feet per minute.

c. Acceleration: 0 to 3g in any direction.

d. Angular velocity: 0 to 140° per second around any axis.

e. Pitch (longitudinal vehicle axis): -31° to +31° from the horizontal.

f. Roll (transverse vehicle axis): -40° to +40° from the horizontal.

The PADS shall indicate when a boundary for vehicle dynamics is approached that affects the PADS operation and requires corrective action.

3.10.2 Initial positioning. The PADS shall include a device that will, from the host vehicle driver position, provide a means of determining the PADS precise position in relation to a predetermined survey control point on the ground.

3.10.3 Offset coordinate and azimuth transfer. Means shall be provided to transfer position coordinates in and out of PADS, from and to a location offset from the host vehicle. Means shall be provided to transfer azimuth from PADS to an orienting line offset from the host vehicle. The minimum range of transfer distances shall be one to sixteen meters. Transfer shall be possible under any visibility conditions greater than 150 meters.

3.10.4 Response time. Warm up and calibration time shall be not greater than 30 minutes when the temperature is between +125° F (+52° C) and -5° F (-20° C), and shall be not greater than 60 minutes between -5° F and -50° F (-46° C). Warm-up and calibration time begins at system start-up and ends when the system is fully initialized and ready for operation.

3.10.5 System initialization. The system shall operate as specified herein when provided with horizontal and vertical position data for a reference position at the beginning of a mission. The PADS shall provide its own azimuth reference.

3.10.6 Operating modes.

3.10.6.1 Open traverse. The system shall operate as specified herein, to include accuracy as specified in 3.10.7, when furnished no information other than that entered during initialization. The system shall accept and utilize position data to limit system error, when provided by the operator at any time during the mission.

3.10.6.2 Closed traverse. The system shall store position, azimuth, time, and an identifier for each station where a position was established. The system shall be capable of storing not less than

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

50 sets of such data. At the end of a mission the system shall adjust the data to minimize error when provided with the known coordinates of the final or some intermediate station.

3.10.6.3 Azimuth transfer. Azimuth transfer shall be accomplished by both direct transfer techniques and inverse computation on two positions, as selected by the operator. The operator shall have a choice between grid azimuth and geodetic azimuth referenced to north, for azimuth display. A non-ambiguous means of selecting the desired type of display shall be provided.

3.10.6.4 Direct transfer. Direct transfer shall be as specified in 3.10.3. The PADS accuracy specified in 3.10.7 includes error attributable to the device for transferring offset coordinates.

3.10.6.5 Two position method. Azimuth transfer shall be accomplished using inverse computations between two positions established by the system. The system shall perform in accordance with 3.10.7 for distances between the two positions, ranging from 100 to 1000 meters.

3.10.7 Accuracy. The system shall provide the accuracies specified in Table I, for both closed and open traverse operations, when operated within a 55 kilometer radius with the center at the (initialization) update point.

TABLE I. System accuracy requirements.

Latitude (degrees)	0 - 65 N/S		65 -75 N/S	
Zero-velocity update interval (minutes)	5.0	10.0	5.0	10.0
Horizontal position (meters-CEP)	4.0	7.0	7.0	10.0
Vertical position (meters-PE)	2.0	3.0	2.0	3.0
Azimuth (mils-PE)	0.4	0.4	0.6	0.6

3.10.8 Operating time. The PADS shall perform as specified herein for 24 continuous hours; which includes the time necessary to prepare the system for operation, as specified in 3.10.4. This requirement does not preclude the operator from extending operating time indefinitely.

3.10.9 Zero-velocity stops. When operating within the vehicle dynamics' boundaries specified in 3.10.1 and the combined system dynamics do not exceed programmed limitations, no more than one stop for every 10 minutes of travel time shall be required to provide a zero-velocity reference. The average stop shall not be greater than 30 seconds and the maximum duration shall not be greater than 60 seconds. The operator shall be able to select a separate five-minute zero velocity stop interval during system initialization, if desired.

3.10.10 Position stops.

3.10.10.1 Position only. The duration of stops to establish horizontal and vertical position shall not be greater than 2 minutes.

3.10.10.2 Position and azimuth. The duration of stops to establish horizontal and vertical positions and azimuth shall not be greater than 15 minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

3.10.11 Area of operation. The PADS shall operate as specified herein in any geographic area of the world except within 15° latitude of the poles.

3.10.12 Blackout. The PADS shall be capable of operation under blackout conditions, during which no part of the system shall be visible beyond 10 meters from the center of the system. The CDU display may be blanked during blackout operations.

3.11 Physical characteristics.

3.11.1 Controls. All operating controls shall be positioned for ease of manipulation and observation and shall be operable by personnel wearing arctic mittens with linings. Critical adjustment controls shall be provided with guards or lock out devices to avoid accidental change in adjustment during operation. The control panel and keyboard illumination shall be adjustable for day, night, and blackout conditions (see 3.10.12).

3.11.2 Power system.

3.11.2.1 Ground vehicle. The PADS shall function as specified herein when provided with power from the standard host vehicle, 24 vdc, power system (see 3.3.1). The system shall include cables and adapters necessary to connect to the host vehicle power source, via the NATO receptacle, where available. PADS power consumption shall not degrade operation of the host vehicle or other on-board systems.

3.11.2.2 Air vehicle. The PADS shall function as specified herein when provided with power from the standard host vehicle power system (see 3.3.2). A separate AC to DC converter may be used in rotary wing aircraft that do not provide 24 to 28 vdc power. The system shall include cables and adapters necessary to connect to the host vehicle power source. PADS power consumption shall not degrade operation of the host vehicle or other on-board systems.

3.11.3 Electrical protection. The PADS shall be provided with no less than the requirements cited in 3.8.5. Further, overload protection for electrical components shall be chosen from, or tailored to, the following specifications:

ARINC 607-1 - Design Guidance for Avionic Equipment.

NFPA 70 - National Electric Code.

SAE ARP 1199 - Selection, Application, and Inspection of Electric Overcurrent Protective Devices (DoD Adopted).

UL 943 - Underwriter's Laboratories Inc.

3.12 Functionality. The PADS shall function as specified herein without mechanical or electrical failures or performance degradation, when subjected to low temperatures, high temperatures, low pressure (altitude), sand and dust, rain, humidity, vibration, and shock, as specified herein (see 3.10).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

3.12.1 Environmental.

3.12.1.1 Operating conditions. The operating ambient temperature range of the PADS shall be -50° F (-46° C), negligible solar energy, to +125° F (+52° C), with solar radiation to 360 Btu per square foot, per hour. The operating altitudes shall be 500 feet below to 15,000 feet above sea level. The maximum operating temperature shall be derated in accordance with Figure 1.

3.12.1.2 Storage conditions (in transit cases). The ambient temperature range of the PADS shall be -50° F (-46° C) for 6 hours to +160° F (71° C) for 4 hours with humidity to 100 percent. The storage altitudes shall be 500 feet below, to 50,000 feet above sea level.

3.12.1.3 Low pressure (altitude). The PADS shall be capable of operation and storage at varying atmospheric pressures (see 3.12.1.1 and 3.12.1.2) and shall be air transportable in its shipping/storage configuration. The PADS shall survive rapid decompression during transport and shall withstand pressure changes during operation.

3.12.1.4 Sand and dust. The PADS shall be capable of withstanding a 12 hour exposure with dust concentrations maintained at 0.06 ± 0.03 grams per cubic foot and the air velocity regulated from 300 ft/min (1.5 m/s) to 1750 ft/min (8.9 m/s) as specified in 4.6.2.5.

3.12.1.5 Rain. The PADS, in its operational configuration, shall be capable of withstanding rain at a rate not less than 4 in/hr (10 cm/hr) with a wind velocity greater than 40 mph (18 m/s) as specified in 4.6.2.6.

3.12.1.6 Humidity. The PADS shall be capable of withstanding the test parameters listed in 4.6.2.7. After testing the PADS shall be capable of operation as defined in 4.6.2.1 at ambient conditions.

3.12.1.7 Fungus resistance. Materials selected by the contractor to be used in the construction of the PADS shall be fungus-inert (see 6.7.6). Exposed parts of the PADS not constructed of fungus-inert materials shall be treated with a moisture and fungus proof varnish or other commercially available preparation. This requirement need not apply to components within hermetically sealed enclosures.

3.12.2 Vibration. The PADS, in its normal mounting configuration, shall be capable of withstanding a sinusoidal cycling time of 3 hours and sweep times of 36 minutes (5-2000-5 Hz), per axis, when tested as specified in 4.6.2.8.

3.12.3 Shock. The PADS shall be capable of withstanding shocks of the following magnitudes:

a) Peak acceleration	40 g’s.	75 g’s.
b) Effective transient duration (T_E)	6 - 9 ms.	3.5-5 ms.
c) Cross over frequency	45 Hz.	80 Hz

In addition the PADS transit case shall withstand being dropped from a height of 24 inches onto each bottom edge.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

3.13 Electromagnetic interference. The PADS shall have design characteristics for the control of electromagnetic interference emission and susceptibility. All electrical and electronic components that produce electromagnetic energy shall be provided with a continuous, low-impedance path to the host vehicle, resulting in an impedance of less than 2.5 milliohms. Suitable jumpers may be used across any non-conductive junctures, like vibration isolators.

3.13.1 Conducted emissions (CE).

3.13.1.1 CE, power leads, 30 Hz to 10 kHz.

3.13.1.1.1 Applicability. For alternating current (ac) applications, this requirement is applicable starting at the second harmonic of the equipment under test (EUT).

3.13.1.1.2 Limits. Conducted emissions on the power leads shall not exceed the applicable values shown on Figure 2 for Army aircraft (including flight line) from 30 Hz to 10 kHz.

3.13.1.2 CE, power leads, 10 kHz to 10 MHz.

3.13.1.2.1 Applicability. This requirement is applicable for all power leads, including returns, that obtain power from other sources not part of the EUT.

3.13.1.2.2 Limits. Conducted emissions on power leads and returns shall not exceed the values in Figure 3 for a frequency scan from 10 kHz to 10 MHz.

3.13.2 Conducted susceptibility (CS).

3.13.2.1 CS, power leads, 30 Hz to 50 kHz.

3.13.2.1.1 Applicability. This requirement is applicable to equipment and subsystem AC and DC input power leads, not including returns. If the EUT is DC operated, this requirement is applicable over the frequency range of 30 Hz to 50 kHz. If the EUT is AC operated, this requirement is applicable starting from the second harmonic of the EUT power frequency and extending to 50 kHz.

3.13.2.1.2 Limits. Conducted susceptibility shall not exceed the values in Figure 4 for a frequency spectrum scan from 30 Hz to 50 kHz. The EUT shall not exhibit any malfunction, degradation of performance, or deviation from specified indications, beyond the tolerances indicated herein when subjected to a test signal with levels specified in Figure 4. In the past, the requirement has been met under the following condition: when the power source specified in MIL-STD-462, adjusted to dissipate 80 watts in a 0.5 ohm load, was unable to develop the required voltage at the EUT power input terminals and the EUT was not susceptible to the output of the signal source.

3.13.2.2 CS, bulk cable injection, 10 kHz to 400 MHz.

3.13.2.2.1 Applicability. This requirement is applicable to all interconnecting cables, including power cables.

3.13.2.2.2 Limits. The EUT shall not exhibit any malfunction, degradation of performance, or deviation from specified indications when subjected to a test signal with calibration levels as shown in Figure 5. The requirements are also met if 115 dBμA are induced in the cable under test and the EUT is not susceptible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

3.13.2.3 CS, bulk cable injection, impulse excitation.

3.13.2.3.1 Applicability. This requirements is applicable to all aircraft interconnecting cables, including power cables.

3.13.2.3.2 Limits. The EUT shall not exhibit any malfunction, degradation or performance, or deviation from specified indications when subjected to a calibrated test signal as specified in Figure 6 at a 30 Hz rate for one minute.

3.13.3 Radiated emissions (RE).

3.13.3.1 RE, magnetic field, 30 Hz to 100 kHz.

3.13.3.1.1 Applicability. This requirement is applicable for radiated emissions from equipment and subsystem enclosures, and all interconnecting cables.

3.13.3.1.2 Limits. Magnetic field emissions shall not be radiated in excess of the levels shown in Figure 7 at the specified distances of 2.75 inches (7 cm) and 19.7 inches (50 cm).

3.13.3.2 RE, electric field, 10 kHz to 18 GHz.

3.13.3.2.1 Applicability This requirement is applicable for radiated emissions from equipment and subsystem enclosures, and all interconnecting cables. The requirement does not apply at the transmitter fundamental frequencies or to radiation from antennas. The requirement shall be 10 kHz to 18 GHz.

3.13.3.2.2 Limits. Radiated emissions from the EUT and all interconnecting cables shall not exceed the values in Figure 8 for a frequency spectrum scan from 10 kHz to 18 GHz. Testing is required up to 1 GHz or 10 times the highest intentionally generated frequency within the instrument, whichever is greater. Measurements beyond 18 GHz are not required.

3.13.4 Radiated susceptibility (RS).

3.13.4.1 RS, electric field, 10 kHz to 40 GHz.

3.13.4.1.1 Applicability. This requirement is applicable to equipment and subsystem enclosures, and all interconnecting cables.

3.13.4.1.2 Limits. The EUT shall not exhibit any malfunction, degradation of performance, or deviation from specified indications when subjected to the radiated electric fields specified in Table III. Up to 30 MHz, the requirement shall be met for vertically polarized fields. Above 30 MHz, the requirement shall be met for both horizontally and vertically polarized fields. Circular polarized fields are not acceptable.

TABLE III. RS Limits.

Frequency Range	Limit Level (Volts/Meter)
10 kHz to 2 MHz	200
2 MHz to 30 MHz	200
30 MHz to 1 GHz	200
1 GHz to 18 GHz	200

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

3.14 Design, materials, and manufacturing processes. Unless otherwise specified, the design, materials, and manufacturing process selection is the prerogative of the contractor as long as all articles submitted to the government fully meet the operating, interface, ownership and support, and operating environment requirements specified.

3.14.1 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment of components and sub-components.

3.15 Treatment and painting. All exposed surfaces shall be treated and painted for corrosion protection. Finish paint coat on all external surfaces shall be semi-gloss, Army green enamel.

3.16 Identification marking. Identification marking of the PADS shall be legible, permanent and sufficient to identify the contents of the transit case(s) and marking requirements as specified in the contract or purchase order (see 6.2) and may include patent information on the PADS and the PADS components.

3.17 Government-loaned property. Unless otherwise specified (see 6.6), the following property, in the quantities indicated, will be loaned by the Government (see 6.5):

Description	Identification	Quantity
Truck, Utility: Cargo/Troop carrier, 1-1/4 ton, 4 x 4, M998 (HMMWV)	M998	As specified (see 6.2)
Aircraft; Rotary Wing, Utility: (UH-60) or (UH-1)	UH- 60 or UH-1	100 hours flying time

Other vehicles listed in 3.3 are available for inspection at various government installations, and interface drawings are available for loan (see 6.2). Arrangements for access to them shall be scheduled through the contracting officer.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. The first article inspection shall be performed on a minimum of two complete PADS (see 3.2 and 6.3). This inspection shall include the examination of 4.4 and the first article inspection tests indicated in Table IV. Nonconformance to any specified requirement, the failure of any test or the presence of one or more defects shall be cause for rejection.

4.3 Conformance inspection. Conformance inspection shall be performed on all PADS production units and shall include the examination of 4.4 and the conformance inspection tests indicated in Table IV. Nonconformance to any specified requirement, the failure of any test or the presence of one or more defects shall be cause for rejection.

4.4 Examination. Each PADS examined, shall be inspected for compliance with the requirements specified in 3.3 through 3.16. Each element of inspection shall encompass all visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

and functional examinations. Nonconformance to any specified requirement or the presence of one or more defects, lessening required efficiency, shall constitute cause for rejection. Any redesign or modification following failure to meet the specified requirements shall receive particular attention for adequacy and suitability.

4.5 Tests.

4.5.1 Test conditions. Unless otherwise specified, all tests shall be performed under shelter at the climatic conditions existing at the place of test. The PADS shall be operated as specified herein without maintenance, other than the scheduled maintenance as established by the maintenance schedule prepared and submitted by the contractor prior to testing. Failure attributable to Government-loaned equipment will not constitute failure of any test specified herein. Before testing, each PADS shall be subjected to a burn-in as specified in 4.5.1.1.

4.5.1.1 Burn-in. Each PADS shall be subjected to pretest conditioning (burn-in) consisting of repeated applications of temperature, vibration, and ON-OFF cycles as indicated in Figure 9, for 48 operating hours. If the PADS fails during burn-in, it shall be repaired and the burn-in continued until not less than 48 operating hours have been accumulated, with the last two cycles failure-free. Substitute test procedures may be used pending approval by the government.

4.5.2 Test schedule. Unless otherwise specified, all applicable tests shall be performed in the order specified in Table IV.

TABLE IV. Test schedule.

First article inspection	Conformance inspection	Test	Test paragraph	Requirement paragraph
X	X	Burn-in	4.5.1.1	3.6.1
X	-	Performance characteristics	4.6.1	3.10
X	-	Accuracy	4.6.1.3.3	3.10.7
X	-	Maintainability	4.6.1.3	3.7
X	-	Functionality	4.6.2.1	3.12
X	-	Reliability	4.6.1.2	3.6
X	-	Low temperature	4.6.2.2	3.12.1.1, 3.12.1.2
X	-	High temperature	4.6.2.3	3.12.1.1, 3.12.1.2
X	-	Low pressure (altitude)	4.6.2.4	3.12.1.3
X	-	Sand and dust	4.6.2.5	3.12.1.4
X	-	Rain	4.6.2.6	3.12.1.5
X	-	Humidity	4.6.2.7	3.12.1.6

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

TABLE IV. Test schedule - Continued.

First article inspection	Conformance inspection	Test	Test paragraph	Requirement paragraph
X	-	Vibration	4.6.2.8	3.12.2
X	-	Shock	4.6.2.9	3.12.3
X	-	Noise	4.6.3	3.9
X	-	Electromagnetic interference	4.6.4	3.13
X	-	Blackout	4.6.5	3.10.12
X	-	Transit case	4.6.6	3.4.8
X	-	Interchangeability	4.6.7	3.4.11
-	X	Conformance acceptance	4.6.8	3.10.7

4.6 Test methods.

4.6.1 Performance.

4.6.1.1 Operability. The PADS, mounted in a host ground vehicle, shall be subjected to six 5-hour tests over a Government-approved test course(s). These operational tests shall be performed using azimuth alignment. Position, elevation, and azimuth shall be established at each survey control point. Azimuth shall be established by direct transfer at 30 percent of the control points and by two-position determination at the remaining 70 percent. Open traverse survey parameters shall be measured at each control point. Closed traverse survey parameters shall be computed at the end of each test. The host vehicle engine shall be turned off for 1 minute at 75-minute intervals during each test. The system shall be shut down at least 1 hour between tests. Nonconformance to the requirements of 3.10 shall constitute failure of this test.

4.6.1.2 Reliability. The PADS shall be tested to determine system reliability, expressed in mean-time between failures (MTBF). Testing shall continue until either an ‘accept’ or ‘reject’ decision is reached. An 'accept' decision is achieved when an MTBF of 2,000 hours at an 80% confidence level is demonstrated. A decision to ‘reject’ shall be invoked when an accumulation of more than four minor failures (see 6.7.7) occurs within 2,000 hours of operation. The occurrence of one or more major failures (see 6.7.8) shall invoke a decision to ‘reject’ the PADS being tested. Data accumulated during operational testing shall be included as part of reliability testing.

4.6.1.2.1 Test procedure. The PADS shall be installed in the host vehicle, attached to vehicle power and tested over an approved operational test course(s). An ON-OFF cycle of 5 hours ON and 1 hour OFF shall be instituted. Two to four PADS shall constitute the reliability sample. Individual units that have been subjected to the environmental test specified in 4.6.2 shall not be selected for reliability testing. Nonconformance to 3.6 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

4.6.1.3 Maintainability.

4.6.1.3.1 Maintenance support evaluation. Maintenance operations required during testing shall be performed to determine conformance to requirements. The inability to maintain the PADS, utilizing the support elements as specified, shall constitute failure of this test. Errors or deficiencies in manuals shall not be considered in assessing the maintenance ratio or MTBF. Nonconformance to 3.7 shall constitute failure of this test.

4.6.1.3.2 Maintenance ratio. The PADS maintenance ratio shall be computed during testing. Nonconformance to 3.7.3 shall constitute failure of this test.

4.6.1.3.3 Accuracy. Data accumulated during operational and reliability testing shall be used in accuracy computations. Positions used to initialize or update the system or to close a traverse shall not be used. Compute the accuracy as follows:

$$CEP = 1.1774 \ x \left[\frac{RMS_n + RMS_e}{2} \right]$$

$$PE_v = .6745 \ RMS_v \qquad RMS = \left[\frac{\sum_{i=1}^{i=N} (X_i)^2}{N} \right]^{1/2}$$

Where:

X	is the linear error between the PADS established position and the reference coordinate for a single, independent measurement.
N	is the total number of measurements in the sample.
CEP	is the Circular Error Probable (expressed in meters).
PE	is the Probable Error (expressed in meters).
RMS	is the Root Mean Square (expressed in mils)
Subscript n	indicates north error.
Subscript e	indicates east error.
Subscript v	indicates vertical error.

Nonconformance to 3.10.7 by more than 2 percent of the individual tests, the individual survey control points, or the individual measurements shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

4.6.2 Environmental.

4.6.2.1 Functionality. Before and after each environmental test, normal operation of the PADS shall be simulated, including operation of all switches, controls, input and output devices, displays, indicators, and azimuth transfer devices. The PADS performance shall be monitored using system displays as much as possible. The PADS shall be operated with the input supply voltage at the normal level, lowest limit, and highest limit. The PADS accuracy of surveying parameters (Northing, Easting, Elevation, and Azimuth) and the PADS warm-up and calibration times shall be recorded. An erroneous reading, a malfunction, or failure to meet specified values for any parameter or the functionally requirements specified in 3.12 shall constitute failure of the test. Inspections shall be performed on samples as specified in Table V.

TABLE V. Inspection designations.

Inspection	One unit	Two units	
		Unit 1	Unit 2
Low Temperature	1	1	
High Temperature	2	2	
Low pressure	3	3	
Sand and Dust	4		1
Rain	5	4	
Humidity	6	5	
Vibration	7		2
Shock	8		3

Should the pre- and post-functional tests meet specifications from one environmental test to the next, the post-functional test may serve as the pre-functional test for the following test.

4.6.2.2 Low temperature. The temperature of the PADS, in its transit case, shall be stabilized at -50° ±5° F (-46° C) and maintained for 6 hours. The rate of temperature change shall not be greater than 6° F/hr. At completion of this storage conditioning period, it shall be removed from its case and the temperature of the PADS shall be adjusted to -50° ±5° F (-46° C) and power applied at that temperature for 6 hours. When the temperature of the PADS returns to ambient, it shall be tested as defined in 4.6.2.1 and examined for these additional defects, any of which shall also constitute failure of this test:

a. Embrittlement, discoloration, cracking or breaking of material.

b. Binding of parts.

c. Stiffening of shock mounts.

4.6.2.3 High temperature. The temperature of the PADS, in its transit case, shall be stabilized at +160° ±5° F (+71° C) and maintained for 4 hours. The rate of temperature change shall not be greater than 6° F/hr. Solar radiation at a rate of 360 Btu/ft^2/hr shall be concurrently applied. At completion of this storage conditioning period, it shall be removed from its case and the temperature of the PADS shall be adjusted to +125° ±5° F (+52° C) and power applied at that

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

temperature for 6 hours. When the temperature of the PADS returns to ambient, it shall be tested as defined in 4.6.2.1 and examined for these additional defects, any of which shall also constitute failure of this test:

a. Embrittlement, discoloration, cracking or breaking of material.

b. Binding of parts.

c. Loss of lubrication.

d. Deterioration of closure and sealing strips.

4.6.2.4 Low pressure (altitude). The PADS shall be tested in a low pressure (altitude) chamber selected by the contractor and approved by the Government. The altitude change rate shall not exceed 2,000 ft/min (10 m/s), with the exception of the rapid/explosive decompression tests, and the PADS tested shall be subjected to each of these subordinate procedures:

a. The test shall be conducted at a simulated altitude of 50,000 ft, with the PADS in its transit case. Atmospheric pressure shall be stabilized and maintained for not less than 1 hour. The PADS shall then be returned to ambient pressure, removed from its transit case and tested as defined in 4.6.2.1.

b. The test shall be conducted at a simulated altitude of 15,000 ft. The atmospheric pressure shall be stabilized and maintained for 6 hours with the PADS operational during the first and last hours of the test. The PADS shall then be returned to ambient pressure and tested as defined in 4.6.2.1.

c. The test shall be conducted from a simulated altitude of 50,000 ft, with the PADS in its transit case. Atmospheric pressure shall be stabilized and the PADS subjected to a near-instantaneous decompression rate, to ambient, of 0.1 second. The test item fails only if rapid decompression will present a hazard to the carrier or to personnel, and need not be tested as defined in 4.6.2.1.

d. The test shall be conducted from a simulated altitude of 50,000 ft, with the PADS in its transit case. Atmospheric pressure shall be stabilized and the PADS subjected to a rapid decompression rate, to ambient, of 60 seconds. The test item fails only if rapid decompression will present a hazard to the carrier or to personnel, and need not be tested as defined in 4.6.2.1.

4.6.2.5 Sand and Dust. The PADS shall be placed in a test facility selected by the contractor and approved by the Government. The PADS shall be tested for 12 hours, total, using the sand and dust composition and particle size distribution specified in 4.6 2.5.1, with concentrations maintained at 0.06 ± 0.03 grams per cubic foot and the air velocity regulated from 300 ft/min (1.5 m/s) to 1750 ft/min (8.9 m/s). The first portion of the test shall be conducted, with the PADS in its transit case, for 6 hours at +160° ±5° F (+71° C). The PADS shall be operated during the second 6 hour portion of the test, which shall be conducted at 73° ±5° F (23° C). Relative humidity shall be controlled to be less than 30 percent throughout the test. The temperature of

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

the PADS shall be returned to ambient and the PADS shall be tested as defined in 4.6.2.1. It shall also be examined for these additional defects, any of which shall also constitute failure of this test:

a. Penetration of seals.

b. Clogging of openings and filters.

4.6.2.5.1 Sand and dust composition. Sand and dust used in testing the PADS shall be composed of 97 to 99 percent (by weight) silicon dioxide (SiO_2). The following particle size distribution shall be used to produce silica flour for the sand and dust testing:

100% shall pass through a 100 mesh screen.

98 ±2% shall pass through a 140 mesh screen.

90 ±2% shall pass through a 200 mesh screen.

75 ±2% shall pass through a 325 mesh screen.

4.6.2.6 Rain.

4.6.2.6.1 Test facility. The PADS shall be tested in a rain chamber or other facility selected by the contractor and approved by the Government. The rain facility shall simulate falling rain at a rate not less than 10 cm/hr (4 in/hr) with a wind velocity greater than 18 m/s (40 mph). The facility temperature shall be uncontrolled but water temperature shall be maintained at 10° C less than that of the test unit. The rain droplets shall have a diametral range, predominantly, between 2 and 4.5 millimeters. The rain shall be dispersed completely over the test item when accompanied by the prescribed wind.

4.6.2.6.2 Wind source. The wind source shall cause the rain to beat directly and uniformly against one side of the test item, with variations up to 45° from the horizontal. Wind velocity shall be maintained at 64 km/hr (40 mph).

4.6.2.6.3 Additives. A water soluble dye, such as fluorescein, may be added to the rain water to aid in locating and analyzing water leaks. No rust or corrosive contaminants shall be imposed on the test item by the test facility.

4.6.2.6.4 Controls. The rainfall rate and wind velocity shall be verified immediately before testing. The wind velocity shall be measured at the intended position of the test item before placement of the test item in the facility.

4.6.2.6.5 Test procedure-blowing rain. The test procedure for the PADS blowing rain test shall include the following steps:

a. With the PADS in the test facility, in its normal operating configuration, the rain shall be started and the rainfall rate adjusted to 10 cm ± 1 cm/hr (4 in/hr). Following a ten minute exposure, power shall be applied to the unit and initialization and self-alignment completed.

b. Initiate the wind at 18 m/s ± 1m/s (40 mph) and maintain for at least 30 minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

c. PADS switches and keyboard shall be manipulated for the last 10 minutes of each 30 minute segment of the test.

d. Rotate the PADS to expose to the rain source another side of the test item that would be exposed to blowing rain during normal deployment.

e. Repeat steps b - d until all possible variations have been accomplished.

f. Collect, measure, and document (in milliliters) any water found inside the protected areas of the item tested.

g. Test the PADS for compliance with 4.6.2.1 and document the test results.

4.6.2.7 Humidity. The PADS shall be placed in a test facility selected by the contractor and approved by the Government. Using a standard aggravated temperature-humidity 24 hour cycle as shown in figure 10, the duration of the humidity test shall be a total of 16 days, with the PADS in its normal operating configuration. Operational checks, as defined in 4.6.2.1, shall be conducted after the first 24 hour period and at least once every 5 days thereafter at regular intervals. At the conclusion of the test period the PADS shall be stabilized at ambient conditions and again tested as defined in 4.6.2.1. It shall also be examined for these additional defects, any of which shall also constitute failure of this test:

a. Binding of parts.

b. Oxidation and/or galvanic corrosion of metals.

c. Deterioration of electrical components.

4.6.2.8 Vibration. The PADS shall be tested using the vibration apparatus and testing procedures described herein, in its normal mounting configuration. The test method selection shall consist, minimally, of a sinusoidal cycling time of 3 hours and sweep times of 36 minutes (5-2000-5 Hz), per axis. The PADS shall be removed from the vibration test apparatus and examined for damage to its configuration and tested as defined in 4.6.2.1. It shall be examined for these additional defects, any of which shall also constitute failure of this test:

a. Binding of parts.

b. Failure of fasteners.

c. Loosening of component parts.

d. Detrimental component material fatigue.

e. Optical misalignment.

4.6.2.9 Shock.

4.6.2.9.1 Preparation for test.

4.6.2.9.2 Test facility. The test facility shall be selected by the contractor and approved by the Government. The shock test apparatus may be free fall, electrodynamic shaker, or any of the other activating types.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

4.6.2.9.3 Calibration. The shock apparatus shall be calibrated for the specified test requirement for each procedure. Two consecutive shock transients shall be successfully applied to the calibration load before the shock test shall be performed on the actual test item.

4.6.2.9.4 Controls. The instrumentation used to measure shock pulses or shock application spectra shall have the following characteristics:

a. Accelerometer.
   1) A transverse sensitivity less than or equal to 5%.
   2) An amplitude linearity within 10%, from 5% to 100% of the peak acceleration amplitude required for testing.
   3) A flat frequency response within ± 10% across the frequency 5 to 2,000 Hz, for procedure 1, 3, and 4 (see 4.6.2.9.6, 4.6.2.9.8, and 4.6.2.9.9).

b. Analysis system.
   1) Shall not allow more than a 5% measurement error into the frequency band of interest (20 Hz to 10 KHz typically).
   2) A filter having linear phase-shift characteristics shall be used, if filters are used to meet the requirement.
   3) Shall have a pass band within one dB across the frequency range specified for the accelerometer.

c. Apparatus.
   1) The shock apparatus shall be capable of producing a transient that meets or exceeds the required spectral or pulse amplitude. For comparability or repeatability, the shock response spectrum of the transient shall not exceed the test conditions by more than 3.5 dB for procedure 1, 3 and 4 (see 4.6.2.9.6, 4.6.2.9.8, and 4.6.2.9.9).

4.6.2.9.5 Pretest checkout. Each PADS shall be pretested at ambient conditions, to provide baseline data. Conduct the checkout as follows:

a. Conduct a complete visual examination of the test item, with special attention to stress areas and document the results.

b. Install the PADS in the test fixture apparatus and perform an operational check.

c. Document the results and, if the PADS operates satisfactorily, proceed to the first test.

4.6.2.9.6 Functional - procedure 1. Secure the PADS, in its operational mode, to the shock test apparatus by its normal mounting. Operation of the unit is not required during testing.

a. Subject the PADS to each test transient, using the following test parameters:
   1) Peak acceleration 40 g's.
   2) Effective transient duration (T_E) 6 - 9 ms.
   3) Cross over frequency 45 Hz.

b. Record sufficient data to prove that the test item met or exceeded required test levels, to minimally include test logs and photographs of the test setup and actual shock transients.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

c. Inspect the components adjacent to shock or vibration isolated sub-assemblies for impact, within the test item.

d. Perform a functional check on the test item and record performance data.

e. Repeat steps a - d three times for each orthogonal axis, in both the positive and negative directions. (If negative and positive acceleration spectra of one transient meet or exceed desired test levels for an axis, they can be counted as two transients, in each direction.)

f. Document the test results.

4.6.2.9.7 Transit drop - procedure 2. Install the PADS in its transit case. Set the test apparatus to accomplish a drop test from a height of 24 inches onto a 2 inch plywood covered concrete floor.

a. Ensure that the entire unit is at or above 24 inches; that the attribute being tested is the leading edge or surface; and drop the test item, once on its bottom face and once on each bottom edge.

b. Document the impact points and perform a visual inspection after each drop, to facilitate any follow-on failure evaluation that may be required.

c. Remove the PADS from its case, visually examine and document the results, following completion of the required drops.

d. Perform a functional check on the PADS and record performance data. Compare with data obtained during pretest checkout.

e. Document the test results.

4.6.2.9.8 Crash hazard - procedure 3. Secure the PADS, or mockup, to the shock test apparatus by its normal mounting. If a mechanically equivalent mockup is used, it will represent the same hazard potential, mass, center of mass and mass moments about the attachment points, as the PADS.

a. Subject the test item to two shocks in each direction, along the three orthogonal axes of the test item, using the following test parameters:

1) Peak acceleration 75 g’s.

2) Effective transient duration (T_E) 3.5-5 ms.

3) Cross over frequency 80 Hz

b. Record sufficient data to prove that the test item met or exceeded required test levels; to minimally include test logs and photographs of the test setup and actual shock transients.

c. Perform a physical inspection of the test item.

d. Document the results of the physical inspection including an assessment of potential hazards created by equipment breakage or deformation.

4.6.2.9.9 Bench handling - procedure 4. Configure the PADS as it would be for normally scheduled servicing and place it on a rigid, flat, horizontal bench top. The PADS shall not be operated during this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

a. Using one edge of the bottom surface as a pivot, lift the opposite edge of the chassis until it forms an angle of 45° to the bench top. Let the chassis drop freely to the bench top.

b. Repeat, using the other edges of that surface as pivot points, for a total of four drops.

c. Perform a visual inspection after each drop, to facilitate any follow-on failure evaluation that may be required.

d. Repeat a - c, from each surface on which the test item would practically be placed during servicing.

e. Visually examine the PADS on completion of required drops. Inspect the adjacent components of shock or vibration isolated sub-assemblies for impact, within the test item.

f. Perform a functional check on the PADS and record performance data. Compare with data obtained during pretest checkout.

g. Document the test results.

4.6.3 Noise level. With the PADS mounted in the host vehicle and operating, an observer stationed within 1.5 meters of the PADS shall conduct radio/telephone conversations and note noise interference emissions from the PADS. Nonconformance to 3.9 shall constitute failure of this test.

4.6.4. Electromagnetic interference. The PADS shall be tested for electromagnetic emissions and susceptibility characteristics. The contractor shall submit, for acceptance by the contracting officer, a description of the PADS electronic components for evaluation to establish the selection of a testing device and plan to measure the electromagnetic emissions being produced by the PADS. This description shall include a proposed test plan. The approved, or a substituted, plan shall be implemented to test completed design. Failure to meet the requirements of 3.13 shall constitute failure of this test.

4.6.5 Blackout. The PADS, installed in the host-vehicle, shall be operated in an open area under blackout conditions. Eight observers shall be equally spaced in a circle around the vehicle at a distance of 50 meters from it. The observers shall move slowly toward the vehicle until the PADS can be visually detected. The distance between the PADS and the position from which the unit is first detected shall be measured. Nonconformance to 3.10.12 shall constitute failure of this test.

4.6.6 Transit cases.

4.6.6.1 Durability. After each environmental test involving the PADS transit case (see 4.6.2.2, 4.6.2.3, 4.6.2.4, 4.6.2.5 and 4.6.2.9), the case shall be opened and closed not less than three times. Nonconformance to 3.4.8.1 shall constitute failure of this test.

4.6.6.2 Survivability. Dummy loads may be substituted to simulate actual contents in weight, shape, and position in case. Packing and special conditioning of the test specimens shall not be permitted. All transit cases shall be dropped 24 inches onto 2 inch plywood, over a concrete floor. The entire edge or surface to be tested shall be at or above the 24 inch point and shall be the leading edge or surface to take the force of the drop. The PADS transit case shall be dropped

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

once on each bottom edge and the bottom face or skids. After drop testing, each case shall be opened and closed at least three times. Nonconformance to 3.4.8.2 shall constitute failure of this test.

4.6.6.3 Leakage (Immersion) test. The closed transit case shall be stabilized at 145° ±4° F (63° C), then submerged in water to a depth of 6 ± 2 inches for not less than 2 hours. The elapsed time between removing the case from the temperature control chamber and its immersion in the water shall not be greater than one minute. The water temperature shall be controlled to 64° ±5° F (18° C). All dunnage, packing, padding material, etc., that may absorb water shall be removed, if possible, from the transit case before testing. A water soluble dye shall be added to the water to aid in locating and analyzing water leaks. Nonconformance to 3.4.8.3 shall constitute failure of this test.

4.6.7 Interchangeability. Components supplied as spare or repair parts shall be substituted for those, with the same part number, they are intended to replace in the PADS. They shall be examined for form and fit, with respect to their interface with mating and adjoining components. Failure to install, or the modification of any component to accommodate, shall constitute failure of this test. Then, normal operation of the PADS shall be simulated, including operation of all switches, controls, input and output devices, displays, indicators, and azimuth transfer devices. The PADS accuracy of surveying parameters (Northing, Easting, Elevation and Azimuth) and the PADS warm-up and calibration times shall be recorded. Any erroneous reading, malfunction, or failure to meet specified values for any parameter or the functionality requirements specified in 3.12 shall also constitute failure of the test.

4.6.8 Conformance acceptance test. Each PADS shall be operated over a closed loop Production Acceptance Course, reviewed and approved by the Government, to simulate normal operating modes, including operation of all switches, controls, input and output devices, displays, indicators, and azimuth transfer devices. The PADS performance shall be monitored using on-board system displays as much as possible. The PADS accuracy (unadjusted and smoothed Northing, Easting, Elevation, and Azimuth) shall be recorded. The PADS warm-up and calibration times shall be recorded. The Production Acceptance Course shall be run twice. The first run shall use a 10 minute zero velocity update. The second run shall use a 5 minute zero velocity update. The Production Acceptance Test Course shall have at least a 15 kilometer north-south leg and a 15 kilometer east-west leg. Required accuracies shall be as stated in 3.10.7 (0-65° N/S latitude). Failure to meet the requirements of 3.10.7 shall constitute failure of this test.

5. PACKAGING

5.1 Packaging. For acquisition purpose, the packaging requirements shall be as specified in the contract or order (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

6. NOTES

(This Section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Position and Azimuth Determining System is intended for use in the conduct of field artillery surveys to determine the Universal Transverse Mercator (UTM) coordinates and elevations of locations critical in the fire control function, such as battery centers, launcher positions, orientating stations, target acquisition devices, and survey control points. The system will determine the azimuth of the orienting lines for weapons and target acquisition systems. The system will operate in any geographic area except within 15° latitude of the poles.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval, and numbers of PADS required (see 3.2).

c. When mounting interface is required and type of host vehicle (3.3).

d. Cables required and type of host vehicle (3.4.7).

e. When maintenance test equipment is required (see 3.7.1).

f. PADS identification and marking required (see 3.16).

g. When Government-loaned property is not required or type and number/hours of vehicles required (see 3.17).

h. When interface drawings are required (see 3.17).

i. Degree of preservation and packing required (see 5.1).

6.3 First article. When a first article inspection is required, the item to be tested should be an initial production sample. The first article should consist of two or more PADS. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article (see 3.2).

6.4 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of the PADS.

6.5 Interface. The contracting officer should indicate in the contract order the specific host vehicle(s) for which a mounting and/or a power interface will be required.

6.6 Government-loaned property. The contracting officer should arrange to lend the property specified in 3.17 when applicable (see 6.2). The rotary wing aircraft identified in 3.3.2 as a PADS interface requirement, and in 3.17 as Government loaned property should be furnished to the contractor for the specified hours of flying time to aid in accomplishing interface and testing requirements. The aircraft, operated by military personnel, will not be transferred to the contractor.

6.7 Definitions. For the purposes of this specification, the following definitions apply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

6.7.1 Interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts, of mating or adjoining parts, except for adjustment, and without selection for fit or performance.

6.7.2. Maintenance ratio. A maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time.

6.7.3 Uncontrolled safety hazard. Uncontrolled safety hazards may be interpreted as defined in *MIL-STD-454, Requirement 1, Safety Design Criteria-Personnel Hazards.*

6.7.4 Electro-Magnetic Interference (EMI). EMI requirements may be found in applicable portions of MIL-STD-462.

6.7.5 Dissimilar metals. For its purpose, this document considers metals dissimilar when two specimens in contact with each other promote accelerated galvanic corrosion.

6.7.6 Recovered materials. For the purpose of this requirement, recovered materials are those that have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials.

6.7.7 Fungus-inert. A material that, in all modified states and grades, is not a nutrient to fungi is considered fungus-inert.

6.7.8 Minor failure. A minor failure is defined as any malfunction that can be corrected within 30 minutes by adjustment, repair, or replacement using controls and on-equipment tools or parts.

6.7.9 Major failure. A major failure is defined as any malfunction that cannot be corrected within 30 minutes by adjustment, repair, or replacement using controls and on-equipment tools or parts, and is a malfunction that causes or may cause one or more of the following:

a. Failure to commence operation

b. Cessation of operation

c. Degradation of performance below the designated levels

d. Damage to the system if operation is continued.

e. Personnel safety hazards.

6.8 Subject term (key words) listing.

Easting

Elevation

Latitude

Longitude

Northing

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

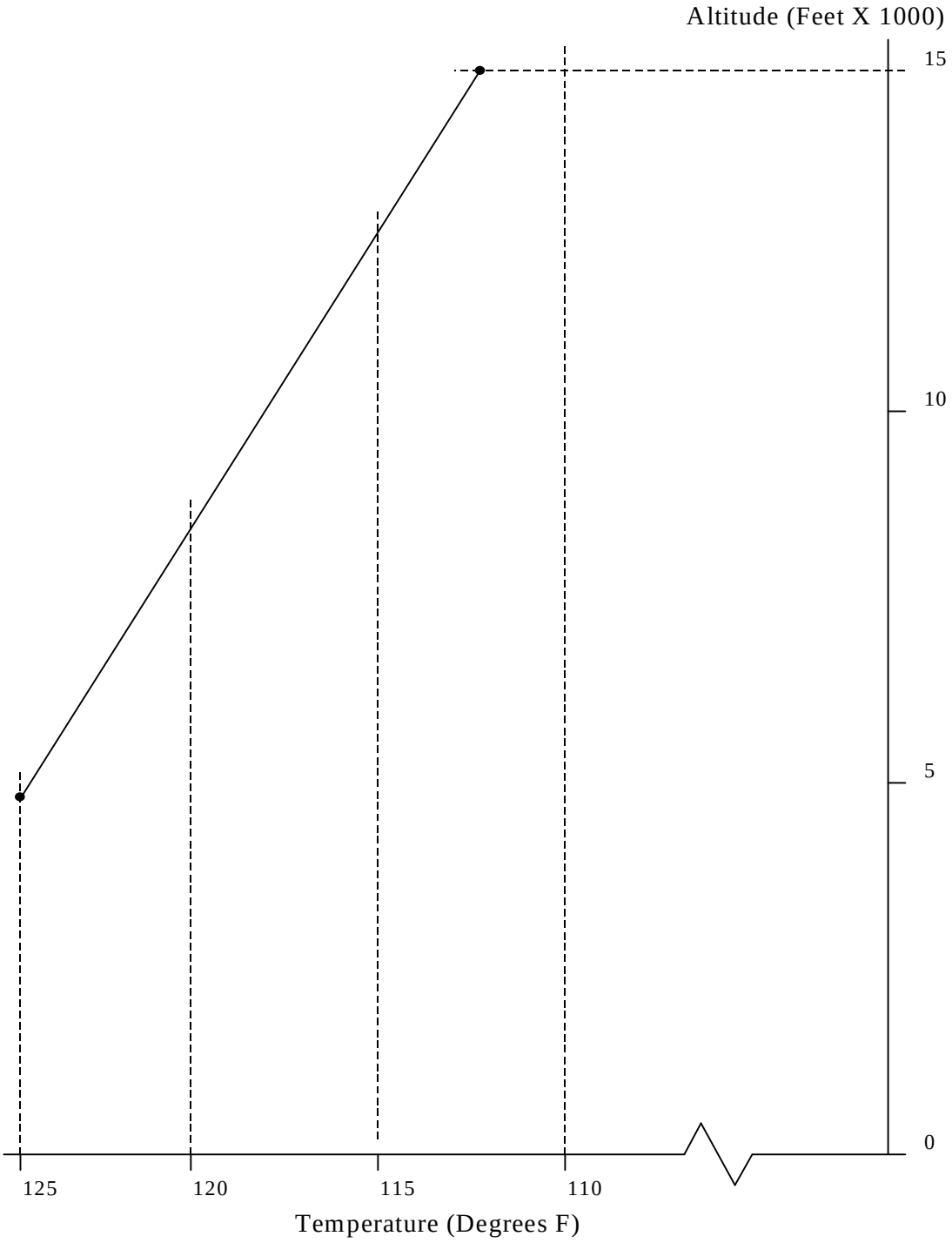


FIGURE 1. Maximum ambient temperature vs altitude.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

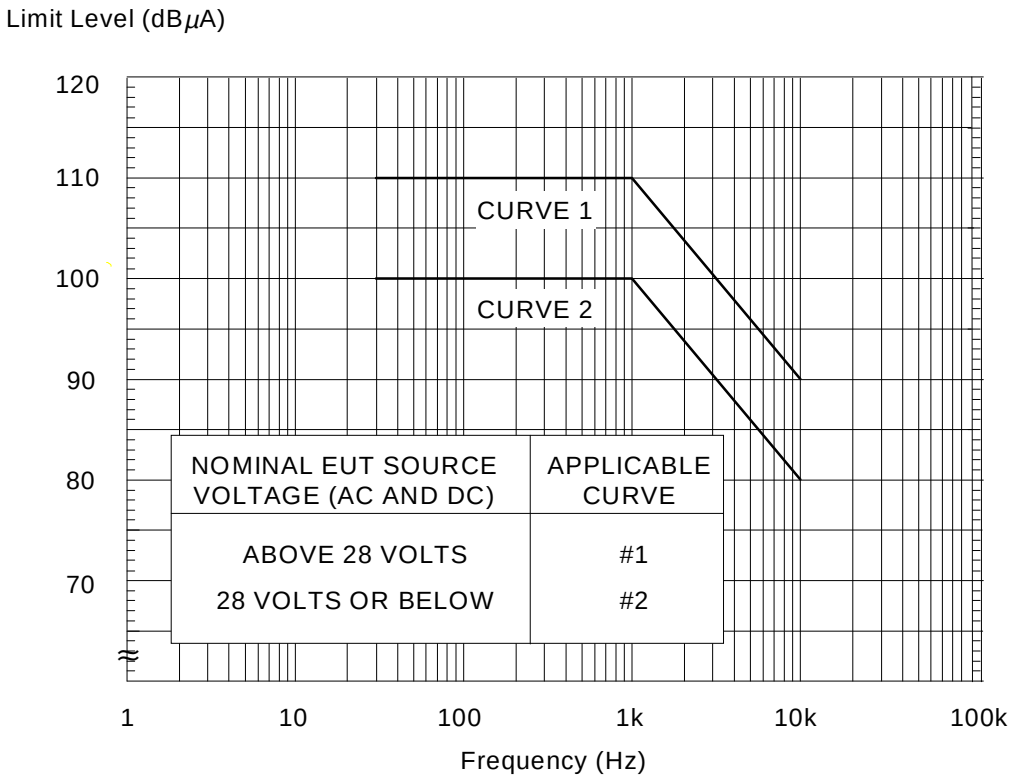


FIGURE 2. CE limits (EUT power leads, ac and dc) for Navy ASW and Army aircraft applications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

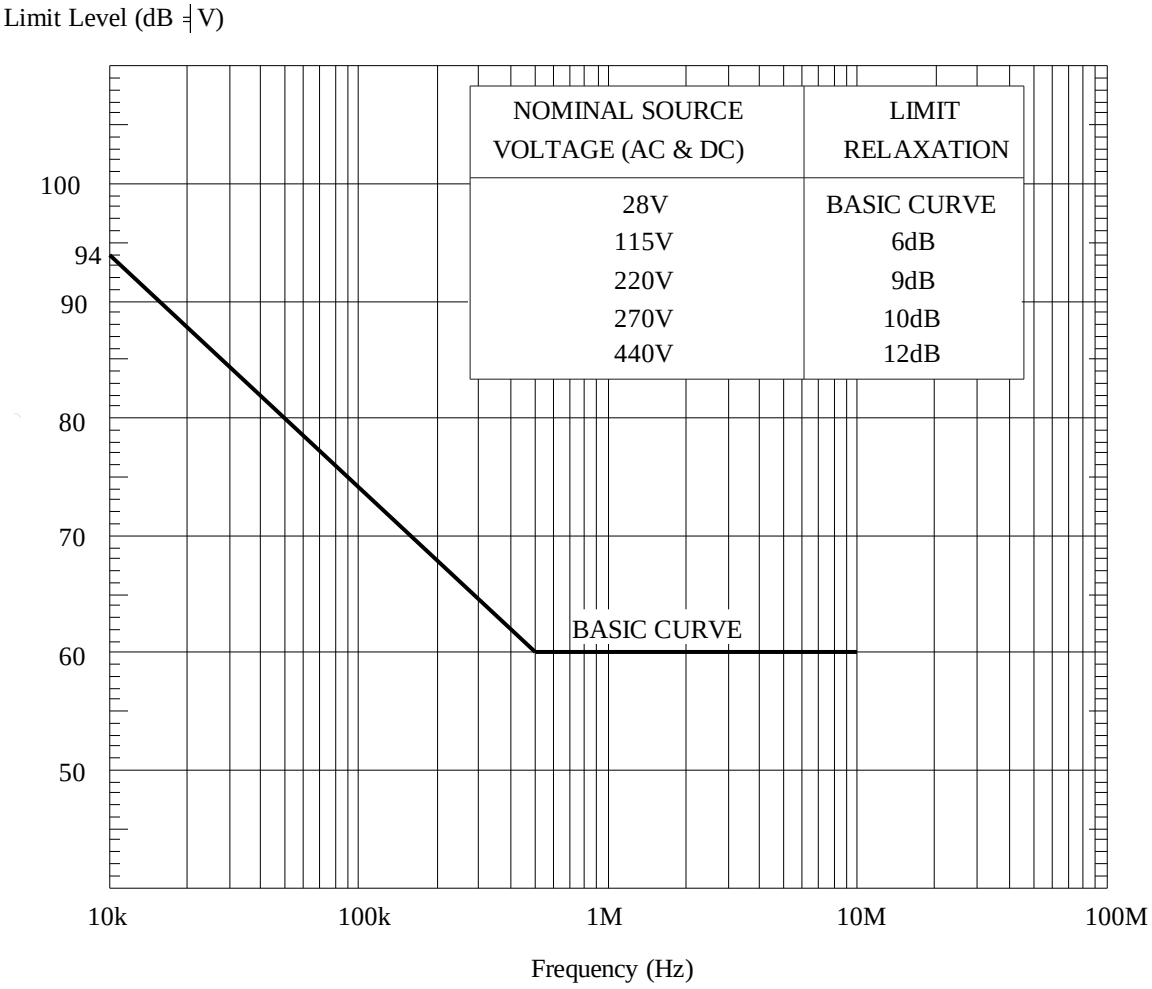


FIGURE 3. CE limits (EUT power leads, AC and DC) for all applications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

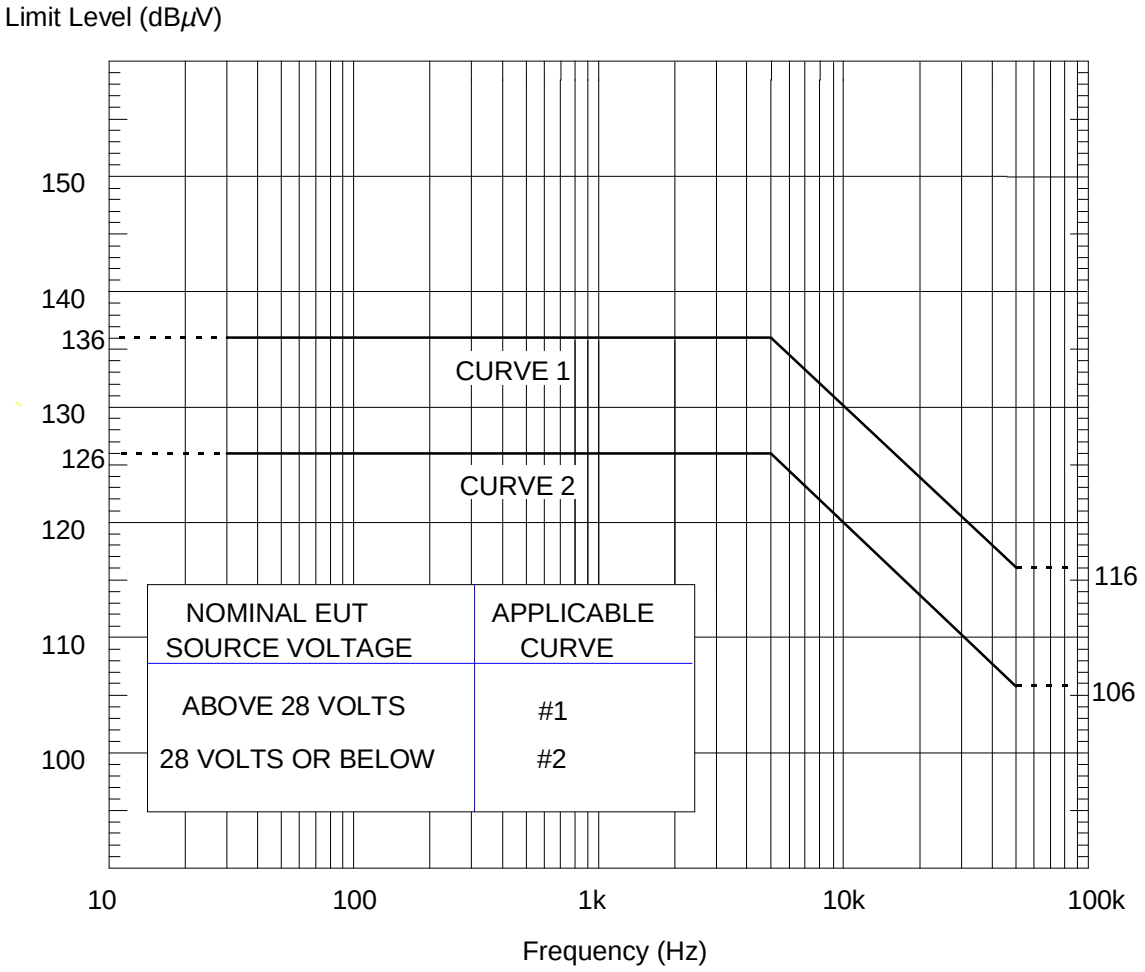


FIGURE 4. CS limits (EUT power leads, AC and DC) for all applications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

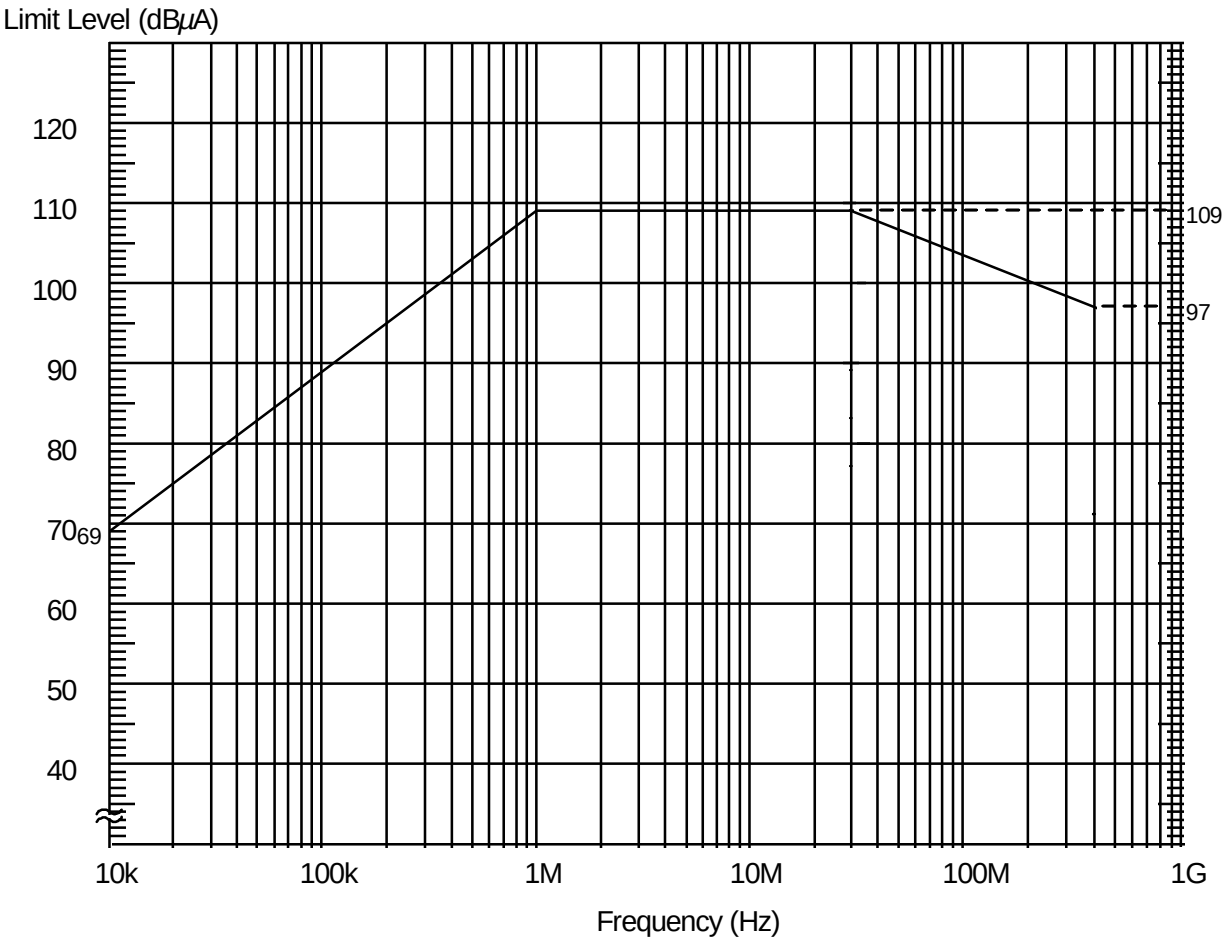


FIGURE 5. CS calibration limits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

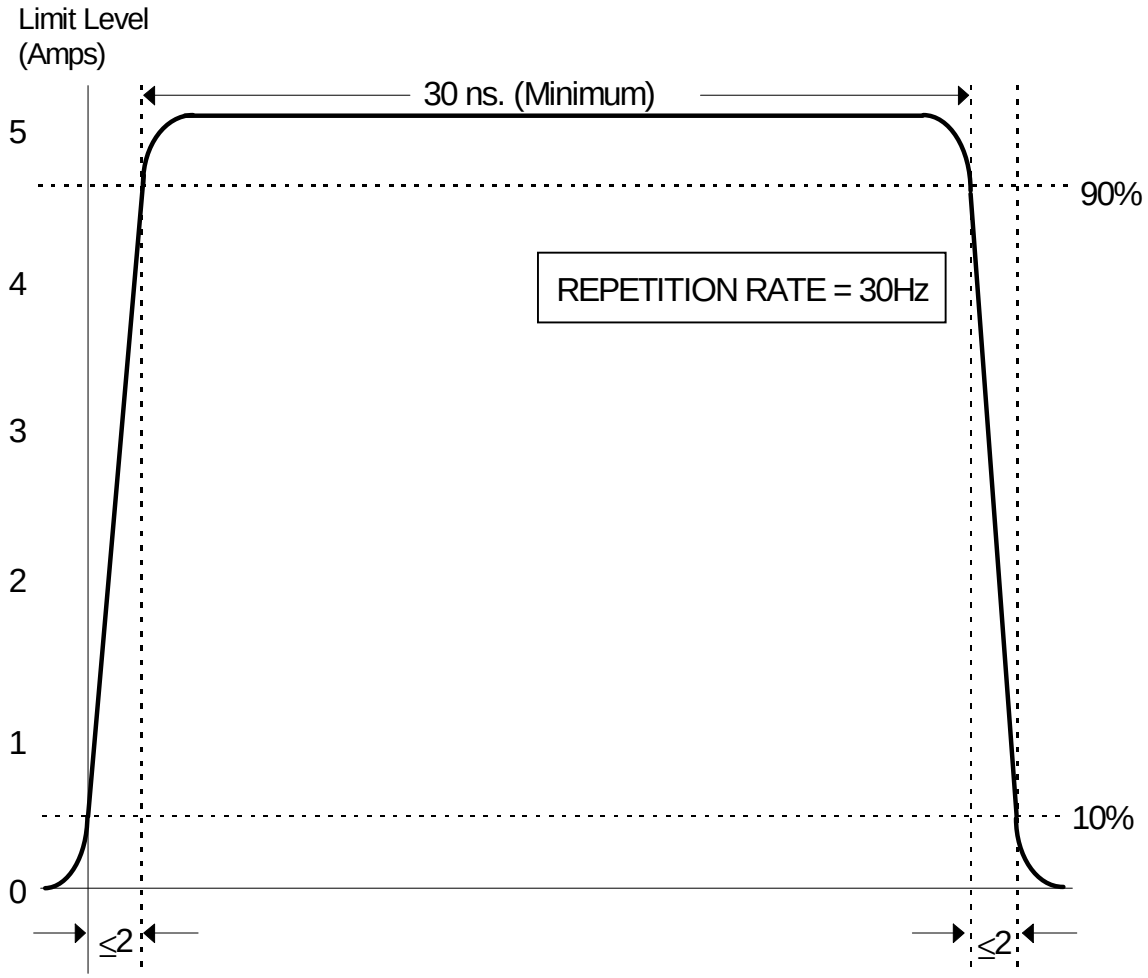


FIGURE 6. CS calibrated signal source characteristics for all applications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

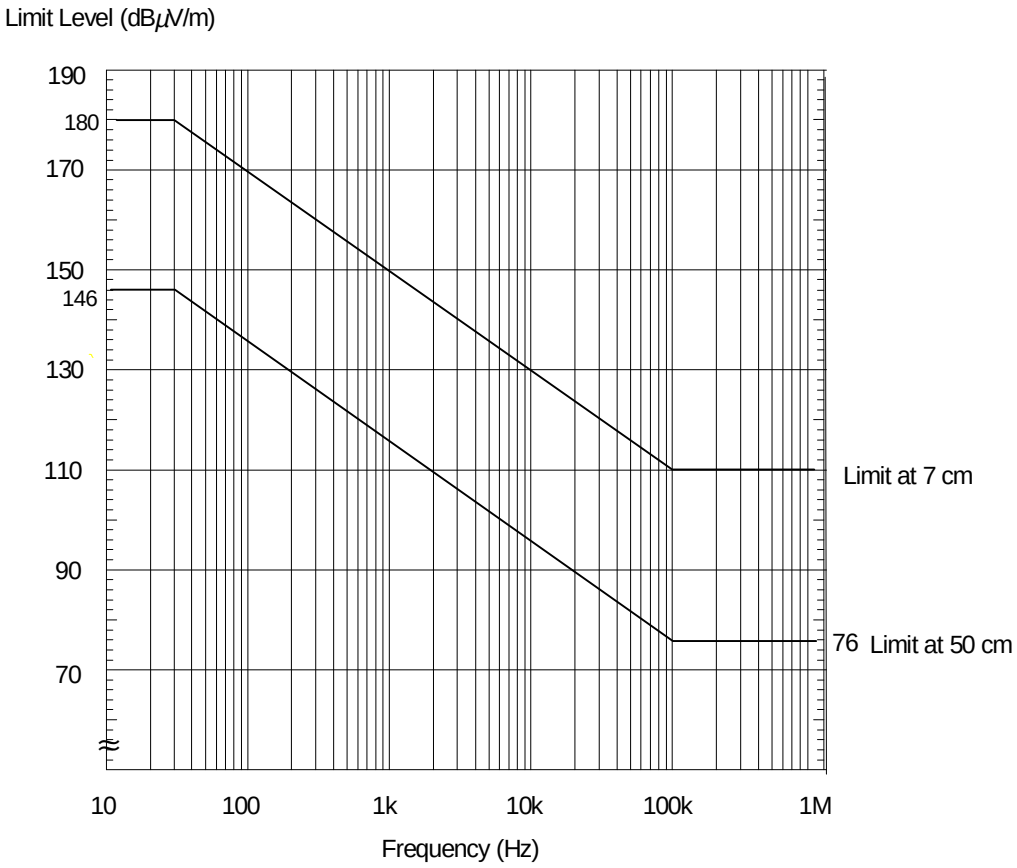


FIGURE 7. RE limits (Army only) for all applications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

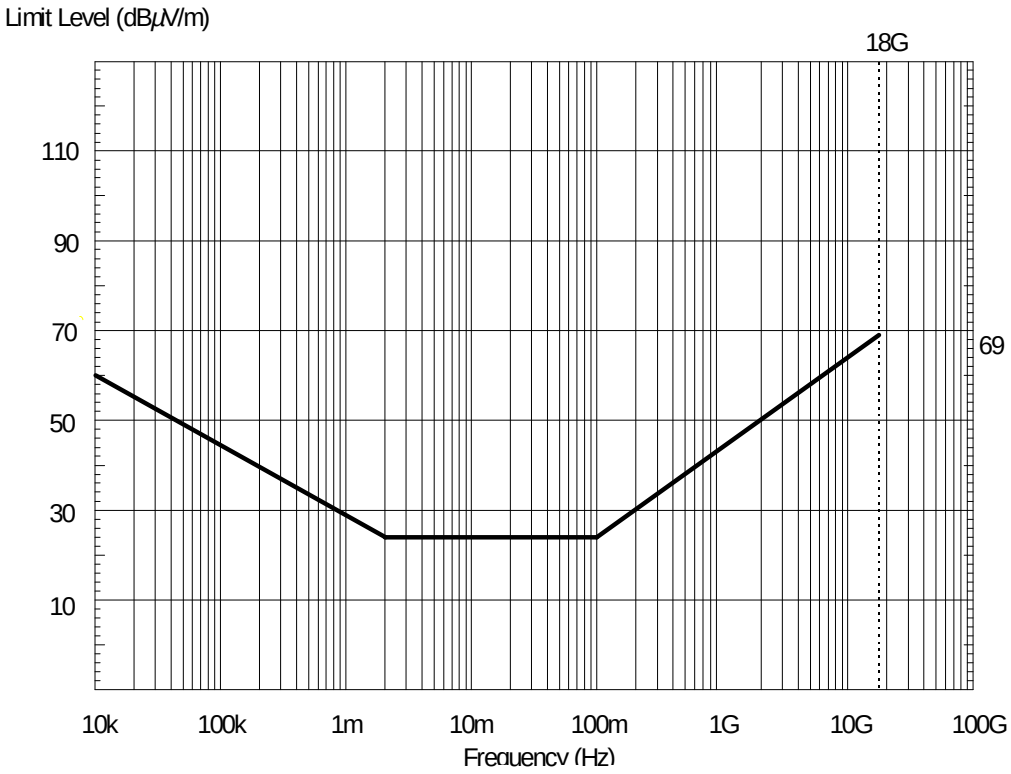


FIGURE 8. RE limits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

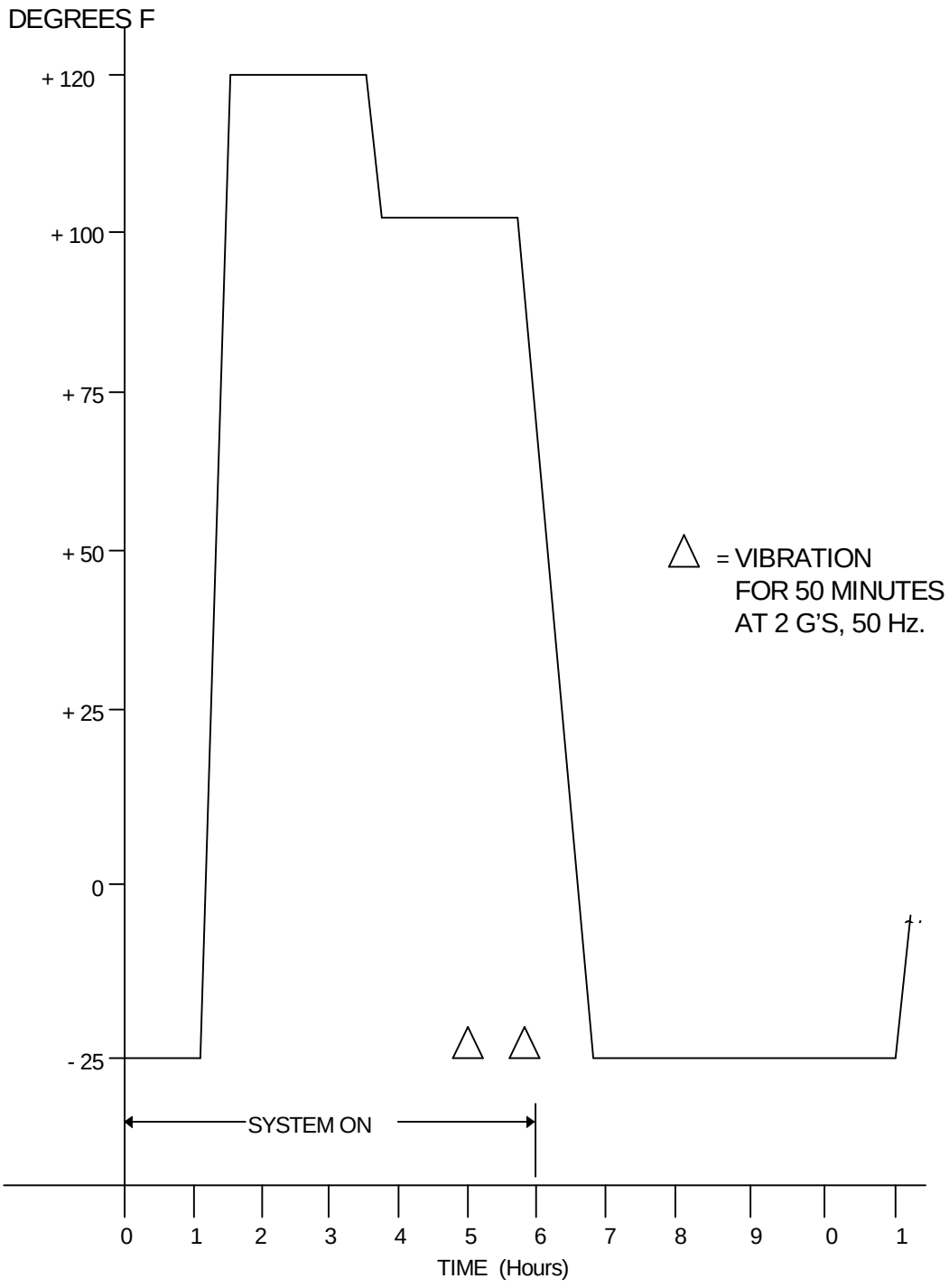
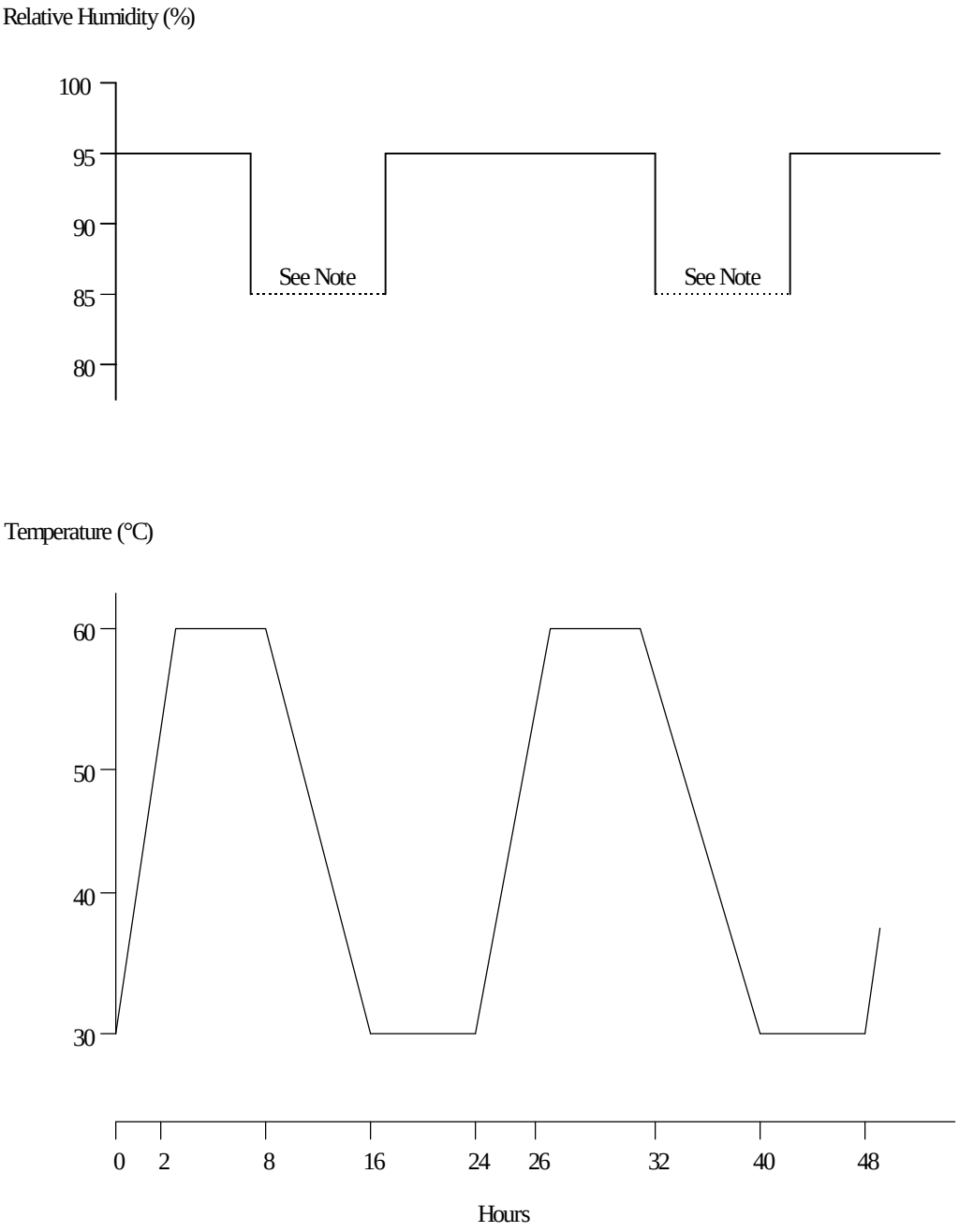


FIGURE 9. Burn-in-cycle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.



Note: Relative humidity maintained above 85% during temperature drops.

FIGURE 10. Aggravated temperature-humidity cycles.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

Custodian:
Army - CE
Navy - MC

Preparing activity
Army - CE

(Project 6675-0010)

Review activity:
DLA - GS
Navy - AS

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter must be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-52955B	2. DOCUMENT DATE (YYMMDD) 961223

3. DOCUMENT TITLE
Position and Azimuth Determining System (PADS)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
C. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME TOPOGRAPHIC ENGINEERING CENTER	b. TELEPHONE (Including Area Code) (1) Commercial (703)428-6862 (2) AUTOVON DSN 328-6862
C. ADDRESS (Inlcude Zip Code) TOPOGRAPHIC ENGINEERING CENTER ATTN: CETEC-PD-TE 7701 TELEGRAPH ROAD ALEXANDRIA, VA 22315-3864	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, Oct 89 *Previous editions obsolete.*

Source: https://assist.dla.mil/ - Downloaded: 2026-10-22T21:42Z
Check the source to verify that this is the current version before use.